1. Integration by parts for indefinite integrals

Recall the definite formula. If u and v have continuous derivatives on $[a, b]$, integrating the product rule $(uv)' = u'v + uv'$ gives

$$\int_a^b u(x)v'(x)\,dx = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x)\,dx.$$

The same product rule tells us how to find an antiderivative on an interval. If $H' = u'v$, then $(uv - H)' = uv'$. Thus

$$\boxed{\int u\,dv = uv - \int v\,du,} \qquad du = u'(x)\,dx, \quad dv = v'(x)\,dx.$$

Choose one antiderivative when evaluating the integral on the right. Add C to the result to obtain all antiderivatives on the interval.

Example

Find $\int x^2 e^x\,dx$. Take $u = x^2$ and $dv = e^x\,dx$. Then $du = 2x\,dx$ and $v = e^x$. The first application lowers the degree of the polynomial:

$$\int x^2 e^x\,dx = x^2 e^x - 2\int xe^x\,dx.$$

Apply parts again, this time differentiating x:

$$\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x + C_1.$$

Substituting and collecting constants gives

$$\int x^2 e^x\,dx = x^2 e^x - 2(xe^x - e^x) + C = e^x(x^2 - 2x + 2) + C.$$

Example

Find $\int e^x \sin x\,dx$. This time differentiating the trigonometric factor twice brings back the original integrand. Let I be an antiderivative of $e^x \sin x$ and J one of $e^x \cos x$. In both applications take $dv = e^x\,dx$. With constants C_1, C_2,

$$\begin{aligned} u = \sin x: &\qquad I = e^x \sin x - J + C_1, \\ u = \cos x: &\qquad J = e^x \cos x + I + C_2. \end{aligned}$$

Substitute the second equation into the first and collect both copies of I:

$$2I = e^x(\sin x - \cos x) + C_1 - C_2, \qquad \boxed{\int e^x \sin x\,dx = \tfrac{1}{2}e^x(\sin x - \cos x) + C.}$$

Differentiating the answer gives $\frac{1}{2}e^x(\sin x - \cos x) + \frac{1}{2}e^x(\cos x + \sin x) = e^x \sin x$.

2. Area between curves

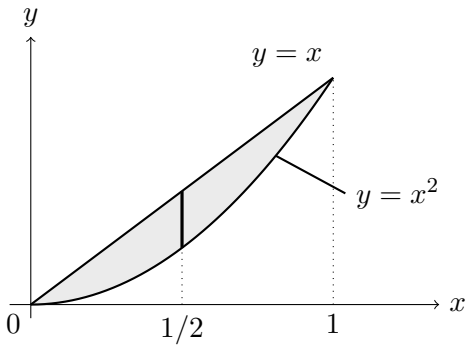
Suppose f and g are continuous on $[a, b]$, with $g(x) \le f(x)$. At each x, a vertical slice runs from $y = g(x)$ to $y = f(x)$. A narrow strip of width Δx_i has area approximately $(f(\xi_i) - g(\xi_i))\Delta x_i$, where ξ_i lies in the strip. Summing the strips and taking the limit gives

$$\boxed{A = \int_a^b (f(x) - g(x))\, dx.}$$

This formula also applies when a graph lies below the x-axis: subtracting the two heights still gives the length of the slice.

Example

Find the area enclosed by $y = x$ and $y = x^2$. From $x = x^2$ the endpoints are $x = 0$ and $x = 1$. The line lies above the parabola between them since $x - x^2 = x(1 - x) \ge 0$. The marked segment at $x = 1/2$ has length $1/2 - 1/4 = 1/4$.



$$A = \int_0^1 (x - x^2)\, dx = \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = \frac{1}{6}.$$

Example

The graphs $y = x$ and $y = x^3$ meet at $x = -1, 0, 1$. On $(-1, 0)$ the cubic is higher, and on $(0, 1)$ the line is higher. Splitting at zero and using symmetry gives the area of both regions:

$$A = \int_{-1}^0 (x^3 - x)\, dx + \int_0^1 (x - x^3)\, dx = 2\left[\frac{x^2}{2} - \frac{x^4}{4}\right]_0^1 = \frac{1}{2}.$$

Integrating $x - x^3$ over the whole interval would give zero because the signed contributions cancel. In general the area between two continuous graphs on $[a, b]$ is $\int_a^b |f(x) - g(x)|\, dx$.

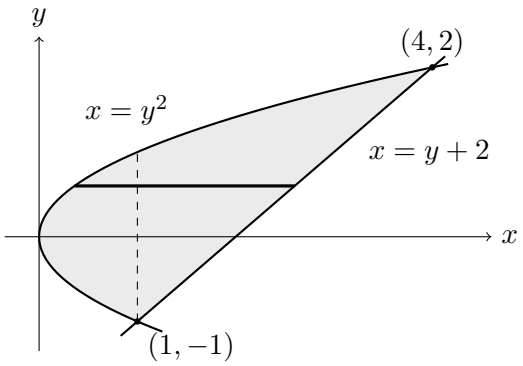
3. Horizontal slices

At height y, suppose a horizontal slice of the region extends from $x = \ell(y)$ to $x = r(y)$. If ℓ, r are continuous and $\ell \le r$ on $[c, d]$, the strip has width $r(y) - \ell(y)$ and thickness Δy. Its contribution leads to

$$\boxed{A = \int_c^d (r(y) - \ell(y))\, dy.}$$

Example

Find the area enclosed by $x = y^2$ and $x = y + 2$. Setting $y^2 = y + 2$ gives $(y - 2)(y + 1) = 0$. The curves meet at $(1, -1)$ and $(4, 2)$.



At each height from $y = -1$ to $y = 2$, the parabola is the left boundary and the line is the right boundary: $(y + 2) - y^2 = (2 - y)(y + 1) \ge 0$. Hence

$$A = \int_{-1}^{2} (y + 2 - y^2)\, dy = \left[\frac{y^2}{2} + 2y - \frac{y^3}{3}\right]_{-1}^{2} = \frac{9}{2}.$$

Vertical slices give a second way to describe this region. The upper endpoint is always $y = \sqrt{x}$. For $0 \le x \le 1$ the lower endpoint is on the lower branch $y = -\sqrt{x}$. For $1 \le x \le 4$ it lies on the line $y = x - 2$. This accounts for the division at the dashed line:

$$A = \int_0^1 2\sqrt{x}\, dx + \int_1^4 (\sqrt{x} - (x - 2))\, dx = \frac{4}{3} + \frac{19}{6} = \frac{9}{2}.$$

4. Volume by cross-sections

Let a solid extend from $x = a$ to $x = b$. Its intersection with the plane $x = t$ is a *cross-section*. Write $S(t)$ for the area of that section. A thin slab between x_{i-1} and x_i is approximated by a prism whose base has area $S(\xi_i)$ and whose thickness is $\Delta x_i = x_i - x_{i-1}$, with $x_{i-1} \le \xi_i \le x_i$. For a solid with continuous cross-sectional area, the resulting volume is

$$\boxed{V = \lim_{\max_i \Delta x_i \to 0} \sum_{i=1}^{n} S(\xi_i)\,\Delta x_i = \int_a^b S(x)\,dx.}$$

For sections perpendicular to the y-axis, integrate their areas with respect to y, from the lowest to the highest section.

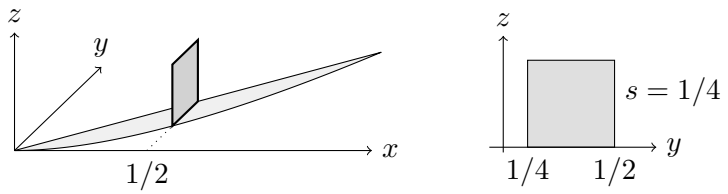
For example, consider a cone whose sections grow as x increases from the tip. Within each slab the smaller end section gives an inscribed cylinder and the larger end section gives a containing cylinder. The difference between the two total cylinder volumes is at most

$$\sum_{i=1}^{n} \big(S(x_i) - S(x_{i-1})\big)\Delta x_i \le \big(\max_i \Delta x_i\big)\big(S(b) - S(a)\big).$$

This tends to zero as the largest thickness tends to zero, so the two sums enclose the same limiting volume.

Example

Use the region $0 \le x \le 1$, $x^2 \le y \le x$ as the base of a solid. In each plane perpendicular to the x-axis the cross-section is a square whose bottom edge runs from $y = x^2$ to $y = x$. Its side is $s(x) = x - x^2$, so its area is $S(x) = (x - x^2)^2$.



The two pictures show the section at $x = 1/2$, first in the solid and then face on. Its side has length $1/4$ and its area is $1/16$. Adding the volumes of all the thin slabs gives

$$\begin{aligned} V &= \int_0^1 (x - x^2)^2\,dx = \int_0^1 (x^2 - 2x^3 + x^4)\,dx \\ &= \left[\frac{x^3}{3} - \frac{x^4}{2} + \frac{x^5}{5}\right]_0^1 = \frac{1}{30}. \end{aligned}$$

For the same base, integrating the side length gave the planar area $1/6$ in Section 2. Here the integrand is the area of a square. Multiplying that area by the slab thickness gives cubic units.